

STATEMENT OF WORK – Super Continuum Laser

1. Objective/Requirements

NASA seeks to procure a short duration, pulsed Super Continuum Laser capable of producing light of wavelength 400nm -1700nm with at least 20nm wavelength bandwidth output. The cumulative total output power of the laser shall be greater than 3W, have a pulse duration of <100ps, and adjustable repetition rate.

2. Characteristics, Scope, and Specifications

- The laser must be capable of lasing light between the wavelengths of 400nm to 1700nm.
- The laser output will have at least a 20nm wavelength bandwidth.
- The cumulative power output will be at least 3W.
- The pulse duration of the laser will be less than 70ps to enable carrier lifetime measurements 1ns.
- The repetition rate shall be adjustable between 0.01 to 40 MHz.
- The laser output should be coupled to an optical fiber cable.
- The laser will be used to measure solar cell bandgap, carrier lifetime, and various other device physics. The laser will use an optical fiber cable to couple into a light-tight enclosure containing the setup for solar cell device characterization.
- The setup is already designed and only the light source is needed.